UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 1 slot(s). Supported theory: 23 slots. Placeholders: 0.
Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 15 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 1 computed, 23 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List two types of malware described in the course and, for each, state one characteristic behaviour that distinguishes it.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Trojans masquerade as helpful or harmless programs while carrying out malicious actions such as stealing data or opening the system to further malware.	"They usually masquerade as helpful or harmless programs and begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed."	051 - Software Security, slide 12	051 - Software Security::s12::1	0.53	supported (ste gate)
2	Worms exploit a vulnerability in the computer system in order to cause harm.	"Worms are another type of malicious software that exploit a vulnerability in the computer system to cause harm."	051 - Software Security, slide 12	051 - Software Security::s12::1	0.88	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 051 - Software Security, slide 12

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six techniques or behaviours by which malware attacks and propagates through computer systems, as described in the course, covering executable infection methods, historical propagation examples, and how specific malware types operate on infected hosts.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A virus can infect an executable by appending its virus body to the program and saving the program's entry point inside the virus body.	"Append a virus body to a program Save an entry point to a program in a virus body"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.64	supported (ste gate)
2	A virus can replace the program's entry point with a jump to the virus body, which later restores the original entry point and jumps to it after executing.	"Virus code restores the original entry point and jumps to it after its own execution"	051 - Software Security, slide 9	051 - Software Security::s09::1	0.61	supported (ste gate)

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3	In EXE infection, the virus overwrites the program header to switch the entry point to a virus and jumps to the original entry point during execution.	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide 10	051 - Software Security::s10::1	0.46 LOW	partly supporte
4	The Morris worm used a debugging feature of sendmail for remote execution and propagated across the Arpanet, penetrating about 6,000 computers.	"Used a debugging feature of sendmail (remote execution)"	051 - Software Security, slide 7	051 - Software Security::s07::1	0.31 LOW	partly supporte
5	Rootkits mask malicious processes and software in infected systems to ensure privileged remote access, and are usually installed after a vulnerability has been found.	"Rootkits are a type of malware that mask malicious processes and software in the infected systems to ensure privileged remote access."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.65	supported (ste gate)
6	Spyware spies on users by stealing logins and passwords, documenting sites visited, logging keystrokes, and redirecting browsers to unwanted sites.	"They steal user logins and passwords, document sites visited, log keystrokes and even redirect browsers to unwanted sites."	051 - Software Security, slide 13	051 - Software Security::s13::1	0.86	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 13; 051 - Software Security, slide 7; 051 - Software Security, slide 9

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how a substitution cipher conceals the original message, and state what serves as the key in such a cipher.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The plaintext letters are swapped for other letters, and the transformed text produced is the ciphertext.	"In substitution ciphers, we replace the plain-text letters with other letters"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s04::1	0.38 LOW	partly supporte
2	The rule governing the letter replacement acts as the key of the cipher.	"The substitution rule is the key"	031 - Classical Cryptography, slide 4	031 - Classical Cryptography::s04::1	0.17 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 031 - Classical Cryptography, slide 4

A2b - 6 marks - computed - matches lecture slide 15 (standard tableau)

Question:

Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.
UMRBVVNWVPREQE

Answer: SECURELOGINNOW

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The 26 x 26 Vigenere tableau is printed in the question paper.

Working:

```
Standard 26 x 26 tableau, A=0; key repeated over the ciphertext.
Ciphertext: UMRBVVNWVPREQE
Key:        CIPHERCIPHERCI
Letter-by-letter: C - K mod 26 = P
  U (20) - C ( 2) = 18 -> S
  M (12) - I ( 8) =  4 -> E
  R (17) - P (15) =  2 -> C
  B ( 1) - H ( 7) = 20 -> U
  V (21) - E ( 4) = 17 -> R
  V (21) - R (17) =  4 -> E
  N (13) - C ( 2) = 11 -> L
  W (22) - I ( 8) = 14 -> O
  V (21) - P (15) =  6 -> G
  P (15) - H ( 7) =  8 -> I
  R (17) - E ( 4) = 13 -> N
  E ( 4) - R (17) = 13 -> N
  Q (16) - C ( 2) = 14 -> O
  E ( 4) - I ( 8) = 22 -> W
Plaintext:  SECURELOGINNOW
```

Suggested marks: 1 mark for the key/table setup; 4 marks for the letter-by-letter working; 1 mark for the complete answer.

Confidence: computed - matches lecture slide 15 (standard tableau).

PART A - QUESTION 3

A3a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State the two stages of the biometric authentication process.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The identification stage, which involves acquisition, creation of master characteristics and storage of master characteristics.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.86	supported (stegate)
2	The authentication stage, which involves acquisition(s), comparison and decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5	072 - Biometrics and Authentication::s05::1	0.83	supported (stegate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

Citations: 072 - Biometrics and Authentication, slide 5

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss six aspects of biometric mechanisms and technologies used for user authentication, covering how biometrics are defined, how they are categorised, and the characteristics of specific biometric technologies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A biometric is defined as a physiological or behavioural characteristic of a human being that can distinguish one person from another and can theoretically be used for identification or verification of identity.	"A biometric is a physiological or behavioural characteristic of a human being that can distinguish one person from another and that theoretically can be used for identification or verification of identity."	072 - Biometrics and Authentication, slide 2	072 - Biometrics and Authentication::s02::1	0.92	supported (stegate)

2	Biometric applications available today are categorised into two sectors, namely psychological and behavioural.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.67	supported (ste gate)
3	The psychological category of biometrics includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.70	supported (ste gate)
4	The behavioural category of biometrics includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4	072 - Biometrics and Authentication::s04::1	0.57	supported (ste gate)
5	Retinal scanning is very efficient and cannot be spoofed, but it suffers from disadvantages such as the user having to look directly, a high failure-to-enrol ratio, and acceptability concerns about light exposure and hygiene.	"Very efficient - cant be spoofed!"	072 - Biometrics and Authentication, slide 9	072 - Biometrics and Authentication::s09::1	0.12 LOW	partly supporte
6	Fingerprint biometrics have disadvantages including racial issues, dirt, grime and wounds, placement of the finger, too large a database to process, and the need for liveness detection because they can be spoofed.	"Can be spoofed - liveness important!"	072 - Biometrics and Authentication, slide 7	072 - Biometrics and Authentication::s07::1	0.13 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 2; 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 7; 072 - Biometrics and Authentication, slide 9

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how the classes of threats described in the course can affect the confidentiality, integrity and availability of information assets.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Disclosure threats such as snooping, sniffing and wiretapping give unauthorised access to information, which harms confidentiality.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.27 LOW	supported
2	Deception and disruption threats involve modification of data, which harms integrity.	"Deception - acceptance of false data"	011 - Introduction to Computer Security, slide 10	011 - Introduction to Computer Security::s10::1	0.25 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 10

A4b - 6 marks - supported by lecture slides (not lecturer-marked)

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Question:

Refer to your answer in (a). Discuss the Samy worm as a case study of script injection, covering how the malicious script is delivered and executed, how the worm spread itself, and the scale of the resulting infection.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The worm injected JavaScript into a user profile using document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.78	supported (ste gate)
2	The worm obtained the list of the user's heroes by issuing a GET request on a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.38 LOW	supported
3	The worm added Samy as a friend by performing XML-HTTP on the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.78	supported (ste gate)
4	Any visitor to an infected user's page had their own profile and friends list infected, so the worm spread from profile to profile.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15	052 - Web Security::s15::1	0.46 LOW	supported
5	The infection eventually reached about 1,000,000 MySpace profiles, showing the scale of the impact.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15	052 - Web Security::s15::1	0.50	supported (ste gate)
6	The underlying weakness is that any web application returning user input without filtering is a potential XSS vulnerability, which is what allowed the injected script to be stored and served.	"Any web application that returns user input without filtering is a potential XSS vulnerability!"	052 - Web Security, slide 13	052 - Web Security::s13::1	0.56	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 052 - Web Security, slide 13; 052 - Web Security, slide 15

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The model was proposed by Brewer and Nash to address conflicts of interest faced by consultants or contractors.	"Brewer and Nash (1989) proposed a policy called the Chinese Wall Policy that addresses such conflicts of interest."	024 - Formal Security Model, slide 21	024 - Formal Security Model::s21::1	0.56	supported (ste gate)
2	Access rights are not fixed but depend dynamically on the subject's history of past	"The Chinese Wall is an access control policy in which accesses are sensitive to the history of past accesses."	024 - Formal Security Model, slide 27	024 - Formal Security Model::s27::1	0.30 LOW	supported

	accesses.					

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 024 - Formal Security Model, slide 21; 024 - Formal Security Model, slide 27

A5b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six (6) points about the access control matrix, access control lists and capabilities as access control structures, covering what each structure represents and one advantage or disadvantage of each.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The access control matrix is identified as a basic access control structure.	"ACM is a basic control structure."	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.60	supported (ste gate)
2	A stated advantage of the access control matrix is that it is easy to verify.	"Advantages: clarity of definition easy to verify"	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.50	supported (ste gate)
3	A stated disadvantage of the access control matrix is poor scalability.	"Disadvantages: poor scalability poor handling of changes"	023 - Access Control and Authorization, slide 8	023 - Access Control and Authorization::s08::1	0.50	supported (ste gate)
4	An access control list is described as an object-centered description of access rights.	"ACL is an object-centered description of access rights:"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.83	supported (ste gate)
5	A stated advantage of ACLs is easy access to object access rights.	"Advantages: easy access to object access rights relative easiness of management using abstractions"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.83	supported (ste gate)
6	A stated disadvantage of ACLs is difficulty of revocation.	"Disadvantages: poor overview of access rights per subject difficulty of revocation difficulty of sharing"	023 - Access Control and Authorization, slide 10	023 - Access Control and Authorization::s10::1	0.50	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 10; 023 - Access Control and Authorization, slide 8

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State two ways in which an intrusion detection system can support the detection and defence against attacks on a computer system.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	An IDS can confirm whether an attack succeeded or failed by examining host-based IDS log entries.	"Verifies success or failure of attack by reviewing HIDS log entries"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.25 LOW	supported
2	An IDS can monitor network traffic even when it is encrypted.	"Can monitor network encrypted traffic"	062 - Intrusion Detection System, slide 14	062 - Intrusion Detection System::s14::1	0.62	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 062 - Intrusion Detection System, slide 14

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A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six threat classes or attack techniques covered in the course and, for each, state one countermeasure or protective mechanism that can be applied against it.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Threats are potential violations of security, distinct from attacks which are actual violations, so countermeasures must guard against possible rather than only realised violations.	"Threats: A threat is a "potential" violation of security"	011 - Introduction to Computer Security, slide 9	011 - Introduction to Computer Security::s09::1	0.27 LOW	partly supporte
2	Software threats include modification of program code, which can be countered by protecting code integrity.	"Modification of program code"	051 - Software Security, slide 2	051 - Software Security::s02::1	0.20 LOW	partly supporte
3	Viruses and self-replicating code form a software threat class that must be defended against.	"Viruses and self-replicating code"	051 - Software Security, slide 2	051 - Software Security::s02::1	0.36 LOW	partly supporte
4	Buffer overflow is a software threat class, and one countermeasure against it is applying software updates/patching.	"Prevention: vulnerability scanning, software updates"	011 - Introduction to Computer Security, slide 19	011 - Introduction to Computer Security::s19::1	0.20 LOW	partly supporte
5	Script injection such as XSS is a software threat class that must be mitigated in web applications.	"Script injection (e.g. XSS)"	051 - Software Security, slide 2	051 - Software Security::s02::1	0.27 LOW	partly supporte
6	SQL injection is a software threat class, and one countermeasure against it is validating and sanitising user input.	"SQL injection"	051 - Software Security, slide 2	051 - Software Security::s02::1	0.18 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 19; 011 - Introduction to Computer Security, slide 9; 051 - Software Security, slide 2

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how the chain of trust works in the Trusted Computing architecture, referring to the role of the TPM and the measurement of each stage.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Each stage of the platform measures and validates the following stage, with the resulting measurements stored in the TPM's Platform Configuration Registers	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.27 LOW	partly supporte

	(PCRs).					
2	The chain begins at the hardware TPM and then proceeds through the software layers (RTM, TPM Software Stack, BIOS, kernel), so that the whole platform ends up verified as being in a trusted state.	"The chain starts with the hardware TPM, then software:"	073 - Trusted Computing, slide 8	073 - Trusted Computing::s08::1	0.28 LOW	partly supporte

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 8

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State six components and functions of the NGSCB architecture, focusing on the Nexus and the Nexus Computing Agent (NCA).

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	NGSCB has two primary system components, namely the Nexus and the Nexus Computing Agent (NCA).	"Two primary system components Nexus Special kernel (core of the trusted operating)"	073 - Trusted Computing, slide 13	073 - Trusted Computing::s13::1	0.29 LOW	partly supporte
2	The Nexus is a special kernel that forms the core of the trusted operating environment.	"Special kernel (core of the trusted operating)"	073 - Trusted Computing, slide 13	073 - Trusted Computing::s13::1	0.62	supported (ste gate)
3	The Nexus is a security kernel that is authenticated on boot.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.80	supported (ste gate)
4	The Nexus authenticates trusted programs.	"Authenticates trusted programs"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.75	supported (ste gate)
5	The Nexus provides the application interface to the TPM.	"Application interface to TPM"	073 - Trusted Computing, slide 14	073 - Trusted Computing::s14::1	0.60	supported (ste gate)
6	The NCA is a trusted software component that runs in trusted mode and communicates with the Nexus.	"Trusted software component Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15	073 - Trusted Computing::s15::1	0.86	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 073 - Trusted Computing, slide 13; 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List TWO (2) password security controls that can be applied to password files and user accounts.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Applying access control to password files	"Access control applied to password files"	022 - Identification and Authentication, slide 12	022 - Identification and Authentication::s12::1	0.80	supported (ste gate)
2	Deactivating accounts	"De-activation of accounts after	022 -	022 -	0.71	supported (ste

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	after multiple failed login attempts	multiple failed logins"	Identification and Authentication, slide 12	Identification and Authentication::s12::1		gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 12

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how single sign-on, Kerberos and smart cards support user authentication in large distributed environments. Your answer should cover the motivation for single sign-on, the solutions available, how Kerberos handles a user login, and where these mechanisms fit among alternative authentication approaches.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Single sign-on is motivated by reducing the need for users to memorize passwords.	"Reduce the need to memorize passwords"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.50	supported (ste gate)
2	Single sign-on also reduces the time users spend typing passwords.	"Reduce the time spent on typing passwords"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.38 LOW	supported
3	Single sign-on lowers the effort of password maintenance.	"Reducing password maintenance effort"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.50	supported (ste gate)
4	Client-server architectures such as Kerberos and Active Directory are one solution for single sign-on.	"Client-server architectures (Kerberos, Active Directory)"	022 - Identification and Authentication, slide 16	022 - Identification and Authentication::s16::1	0.60	supported (ste gate)
5	In Kerberos login, the user enters a user name and password at a client workstation.	"User enters his user name, U and password, P at a client workstation."	022 - Identification and Authentication, slide 19	022 - Identification and Authentication::s19::1	0.71	supported (ste gate)
6	The hash of the user's password becomes the secret key of the client or user.	"The hash value of his password KU = h(P) becomes a secret key of the client/user."	022 - Identification and Authentication, slide 19	022 - Identification and Authentication::s19::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 16; 022 - Identification and Authentication, slide 19

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the difference between morals, ethics and laws as three distinct forms of guidance on conduct.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Morals are described as traditions of belief concerning right and wrong conduct, forming the foundation of social behaviour.	"Morals are traditions of belief about right and wrong conduct and "doing what is morally right" is the bedrock of our social behavior"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	0.64	supported (ste gate)

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2	Laws are formal rules of conduct imposed by a sovereign authority such as a government on its subjects or citizens.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5	071 - Ethics and Computer Security::s05::1	1.00	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 5

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six ways in which a professional code of ethics and conduct, such as the ACM code, guides the behaviour of a computing professional.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A code of ethics is a documented set of acceptable behaviours for a professional group.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	0.62	supported (ste gate)
2	The code is defined by the professional group itself.	"Defined by the group."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	0.40 LOW	supported
3	It identifies which specific actions are appropriate or inappropriate.	"Identifies specific actions as appropriate or inappropriate."	071 - Ethics and Computer Security, slide 6	071 - Ethics and Computer Security::s06::1	1.00	supported (ste gate)
4	It requires the professional to contribute to society and human well-being, treating all people as stakeholders.	"Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.70	supported (ste gate)
5	It obliges the professional to respect privacy and honour confidentiality.	"Respect privacy."	071 - Ethics and Computer Security, slide 11	071 - Ethics and Computer Security::s11::1	0.33 LOW	partly supporte
6	It directs the professional to give comprehensive and thorough evaluations of computer systems and their impacts, including possible risks.	"Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks."	071 - Ethics and Computer Security, slide 12	071 - Ethics and Computer Security::s12::1	0.82	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 12; 071 - Ethics and Computer Security, slide 6

PART A - QUESTION 10

A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) typical locations where IDS sensors are placed in a network-based IDS architecture.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A sensor can be positioned immediately inside the firewall.	"Just inside the firewall"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.33 LOW	supported
2	A sensor can be positioned on the DMZ.	"On the DMZ"	062 - Intrusion Detection System, slide 7	062 - Intrusion Detection System::s07::1	0.25 LOW	supported

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 062 - Intrusion Detection System, slide 7

A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss the preventive network security mechanisms that an organization can deploy to protect its network, covering firewalls, proxy servers, network segmentation, remote access VPN and intrusion detection/prevention systems. Provide six distinct points.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	A firewall controls incoming and outgoing network traffic according to predetermined security policies.	"Control incoming and outgoing traffic on network with predetermined security policies."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.80	supported (ste gate)
2	A proxy server improves data security and network performance by acting as a firewall and web filter, sharing connections and caching data.	"Proxy servers act as a firewall and web filter, provide shared network connections, and cache data to speed up common requests."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.79	supported (ste gate)
3	Network segmentation creates boundaries between segments whose computing assets share a common function, risk or role.	"Boundaries between network segments where computing assets within the segment have a common function, risk or role within an organization."	061 - Network Security, slide 3	061 - Network Security::s03::1	0.77	supported (ste gate)
4	A remote access VPN gives individual hosts or clients remote and secure access to the organization network.	"Provides remote and secure access to an organization network to individual hosts or clients"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.89	supported (ste gate)
5	Zero Trust Network Access restricts each user to only the access and permission required to fulfill the role.	"A user should only has access and permission that required to fulfill the role"	061 - Network Security, slide 4	061 - Network Security::s04::1	0.64	supported (ste gate)
6	IDS/IPS detect or prevent network attacks such as brute force attacks, Denial-of-Service attacks and exploits of network vulnerabilities.	"To detect or prevent the network security attacks such as brute force attacks, Denial-of-Service attacks and exploits of network vulnerabilities."	061 - Network Security, slide 4	061 - Network Security::s04::1	0.85	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 061 - Network Security, slide 3; 061 - Network Security, slide 4

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss the ethical impacts of technology by examining how professional codes of ethics, such as the ACM code, guide a computing professional's conduct, and evaluate the limitations of such codes in addressing the ethical impacts of technology.

Marking points:

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No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Recognise that the ACM, founded in 1947, has a code of ethics and professional conduct that professionals can refer to when judging the ethical impact of their technological work.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.50	supported (ste gate)
2	Note that other professional bodies such as AITP provide education and forums that help members reason about relevant IT issues and their ethical implications.	"provides quality IT-related education, information on relevant IT issues, and forums for networking with experienced peers and other IT professionals."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.41 LOW	partly supporte
3	Acknowledge that the IEEE Computer Society is another professional body that maintains a code relevant to the ethical conduct of computing professionals.	"Computer Society of the Institute of Electrical and Electronics Engineers (IEEE - CS) (founded in 1946)."	071 - Ethics and Computer Security, slide 7	071 - Ethics and Computer Security::s07::1	0.25 LOW	partly supporte
4	Explain that a professional code may not reflect the ethics of every member of the group, so it cannot fully capture the diverse ethical impacts of technology on different stakeholders.	"May not reflect the ethical of every member of the group."	071 - Ethics and Computer Security, slide 9	071 - Ethics and Computer Security::s09::1	0.44 LOW	supported
5	Discuss that such codes are often not enforceable because they are codes of conduct rather than legal documents, limiting their power to regulate the ethical impacts of technology.	"Many times it isn't enforceable. It's a code of conduct, not a legal document."	071 - Ethics and Computer Security, slide 9	071 - Ethics and Computer Security::s09::1	0.33 LOW	supported

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 7; 071 - Ethics and Computer Security, slide 9

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain five ethical responsibilities that the hospital's information technology unit owes to patients with respect to the privacy and accuracy of their medical records, drawing on the ethical dimensions of computer security.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	The unit must safeguard patients' personal information as a matter of privacy.	"Protecting one's personal information"	071 - Ethics and Computer Security, slide 13	071 - Ethics and Computer Security::s13::1	0.25 LOW	supported
2	The unit must ensure that stored patient information is accurate and can be verified as faithful.	"Deals with authentication and fidelity of information"	071 - Ethics and Computer Security, slide 13	071 - Ethics and Computer Security::s13::1	0.10 LOW	supported
3	The unit must respect who owns information about patients and	"Deals with who owns information about individuals and how information can be sold and	071 - Ethics and Computer Security, slide	071 - Ethics and Computer Security::s13::1	0.55	supported (ste gate)

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	control how that information is sold or exchanged.	exchanged"	13			
4	The unit must govern what information a person is entitled to obtain about others and how that information may be used.	"Deals with what information a person has the right to obtain about others and how the information can be used"	071 - Ethics and Computer Security, slide 13	071 - Ethics and Computer Security::s13::1	0.50	supported (ste gate)
5	Information specialists in the unit carry definite duties to act within ethical, moral and legal limits.	"Users in general, and information specialists in particular, have definite responsibilities in terms of performing within ethical, moral, and legal constraints"	071 - Ethics and Computer Security, slide 3	071 - Ethics and Computer Security::s03::1	0.58	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 13; 071 - Ethics and Computer Security, slide 3

PART B - QUESTION 2

B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

State FIVE (5) points about software insecurity and buffer overflows, covering their root causes, how a stack buffer overflow hijacks execution, and techniques used to strengthen software security.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Software insecurity arises because attackers can modify system resources that are critical for program execution, such as the instruction pointer and function call addresses.	"Software insecurity stems from attacker's ability to modify system resources criticial for program execution, e.g. instruction pointer, function call addresses, interrupt addresses, etc."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.72	supported (ste gate)
2	A key source of software insecurity is the failure to validate user input.	"One of the key sources for software insecurity is failed validation of user input."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.86	supported (ste gate)
3	Buffer overflows are among the most widely used exploitation techniques.	"Buffer overflows are a most widely used exploitation technique."	051 - Software Security, slide 30	051 - Software Security::s30::1	0.71	supported (ste gate)
4	Without proper bound checking, a local buffer can overspill into the adjacent upper stack area.	"Without proper bound checking a buffer content can overspill into adjacent upper stack area."	051 - Software Security, slide 21	051 - Software Security::s21::1	0.92	supported (ste gate)
5	By controlling the buffer content, an attacker can overwrite the return address with an arbitrary value and thereby hijack the execution flow.	"By controlling buffer content, an attacker can overwrite the return address with an arbitrary value and hijack the execution flow."	051 - Software Security, slide 21	051 - Software Security::s21::1	0.86	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 051 - Software Security, slide 21; 051 - Software Security, slide 30

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Windows provides several security mechanisms for managing users, objects and access. Explain FIVE (5)

points about these Windows security mechanisms, covering the directory service used to organise resources, the access rights model, the role of the registry, and why the complexity of these mechanisms can itself be a weakness.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Chunk	Stem overlap	Entailment
1	Active Directory is an LDAP-like directory service introduced in Windows 2000 that organises system resources such as users, groups, credentials, policies and trust.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6	041 - Windows Security::s06::1	0.59	supported (ste gate)
2	Access rights describe what a subject can do to an object and are encoded as a 32-bit mask, with standard rights common to most object types and object-specific rights tailored to each class.	"Encoded as a 32-bit mask"	041 - Windows Security, slide 16	041 - Windows Security::s16::1	0.23 LOW	partly supporte
3	The registry is a hierarchical database holding critical system information, and certain hives such as SAM and Security store security-relevant data.	"A hierarchical database containing critical system information"	041 - Windows Security, slide 4	041 - Windows Security::s04::1	0.33 LOW	partly supporte
4	Windows contains complex mechanisms for user account and object management and access control, and discretionary access control enables fine-grained control over heterogeneous entities.	"Windows systems contain complex security mechanisms for user account and object management as well as access control."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.59	supported (ste gate)
5	The complexity of Windows security mechanisms is also its enemy, because misconfiguration and implementation bugs can lead to severe security incidents.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21	041 - Windows Security::s21::1	0.79	supported (ste gate)

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 041 - Windows Security, slide 16; 041 - Windows Security, slide 21; 041 - Windows Security, slide 4; 041 - Windows Security, slide 6

End of marking scheme.

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